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Trabajo Práctico N° 4: Control Óptimo en Tiempo Discreto.

Ejercicio 1.

Maximizar J= $\sum_{k=0}^{3} 1 + x\ (k) - [u\ (k)]^2$

sujeto a

x (k + 1)= x (k) + u (k),
x (0)= 0,
u (k) ∈ ℝ*, para k= 0, 1, 2, 3.*

N= 4.

F (x, u, k)= 1 + x - u^2.

Final:

$J_4^*\ (x_4)$= S (x (4))
$J_4^*\ (x_4)$= 0.

Período k= 3:

$J_3^*\ (x_3)= \max_{u(3)\in\mathbb{R}} 1 + x\ (3) - [u\ (3)]^2 + J_4^*\ (x_4)$
$J_3^*\ (x_3)= \max_{u(3)\in\mathbb{R}} 1 + x\ (3) - [u\ (3)]^2 + 0$
$J_3^*\ (x_3)= \max_{u(3)\in\mathbb{R}} 1 + x\ (3) - [u\ (3)]^2$.

u^* (3)= 0.

x (3)= x (2) + u (2).

$J_3^*\ (x_3)$= 1 + x (3)
$J_3^*\ (x_3)$= 1 + x (2) + u (2).

Período k= 2:

$J_2^*\ (x_2)= \max_{u(2)\in\mathbb{R}} 1 + x\ (2) - [u\ (2)]^2 + J_3^*\ (x_3)$
$J_2^*\ (x_2)= \max_{u(2)\in\mathbb{R}} 1 + x\ (2) - [u\ (2)]^2 + 1 + x\ (2) + u\ (2)$
$J_2^*\ (x_2)= \max_{u(2)\in\mathbb{R}} 2 + 2\ x\ (2) + u\ (2) - [u\ (2)]^2$.

$\frac{\partial J_2^*}{\partial u(2)}= 0$
1 - 2 u (2)= 0
2 u (2)= 1

u^* (2)= $\frac{1}{2}$.

$\frac{\partial^2 J_2^*}{\partial [u(2)]^2}$= -2 $<$ 0.

x (2)= x (1) + u (1).

J_2^* (x_2)= 2 + 2 x (2) + $\frac{1}{2}$ - $(\frac{1}{2})^2$
J_2^* (x_2)= 2 + 2 [x (1) + u (1)] + $\frac{1}{2}$ - $\frac{1}{4}$
J_2^* (x_2)= $\frac{9}{4}$ + 2 x (1) + 2 u (1).

<u>Período k= 1:</u>

J_1^* (x_1)= $\max_{u(1) \in \mathbb{R}} 1 + x\ (1) - [u\ (1)]^2 + J_2^*\ (x_2)$
J_1^* (x_1)= $\max_{u(1) \in \mathbb{R}} 1 + x\ (1) - [u\ (1)]^2 + \frac{9}{4} + 2\ x\ (1) + 2\ u\ (1)$
J_1^* (x_1)= $\max_{u(1) \in \mathbb{R}} \frac{13}{4} + 3\ x\ (1) + 2\ u\ (1) - [u\ (1)]^2$.

$\frac{\partial J_1^*}{\partial u(1)}$= 0
2 - 2 u (1)= 0
2 u (1)= 2
u (1)= $\frac{2}{2}$
u^* (1)= 1.

$\frac{\partial^2 J_1^*}{\partial [u(1)]^2}$= -2 $<$ 0.

x (1)= x (0) + u (0)
x (1)= 0 + u (0)
x (1)= u (0).

J_1^* (x_1)= $\frac{13}{4}$ + 3 x (1) + 2 * 1 - 1^2
J_1^* (x_1)= $\frac{13}{4}$ + 3 u (0) + 2 - 1
J_1^* (x_1)= $\frac{17}{4}$ + 3 u (0).

<u>Período k= 0:</u>

J_0^* (x_0)= $\max_{u(0) \in \mathbb{R}} 1 + x\ (0) - [u\ (0)]^2 + J_1^*\ (x_1)$
J_0^* (x_0)= $\max_{u(0) \in \mathbb{R}} 1 + 0 - [u\ (0)]^2 + \frac{17}{4} + 3\ u\ (0)$
J_0^* (x_0)= $\max_{u(0) \in \mathbb{R}} \frac{21}{4} + 3\ u\ (0) - [u\ (0)]^2$.

$\frac{\partial J_0^*}{\partial u(0)}$= 0
3 - 2 u (0)= 0

2 u (0)= 3
u^* (0)= $\frac{3}{2}$.

$\frac{\partial^2 J_0^*}{\partial [u(0)]^2}$= -2 $<$ 0.

J_0^* (x_0)= $\frac{21}{4}$ + 3 $\frac{3}{2}$ - $(\frac{3}{2})^2$
J_0^* (x_0)= $\frac{21}{4}$ + $\frac{9}{2}$ - $\frac{9}{4}$
J_0^* (x_0)= $\frac{15}{2}$. Valor óptimo del funcional objetivo.

Por lo tanto, se obtienen los siguientes valores óptimos:

J_0^* (x_0)= $\frac{15}{2}$;
u^* (0)= $\frac{3}{2}$; u^* (1)= 1; u^* (2)= $\frac{1}{2}$; u^* (3)= 0;
x^* (0)= 0; x^* (1)= $\frac{3}{2}$; x^* (2)= $\frac{5}{2}$; x^* (3)= 3.

Ejercicio 2.

Resolver el siguiente problema:

Minimizar J= $\sum_{k=0}^{2}[u\ (k)]^2 + [x\ (k) - 2k]^2 + [x\ (3)]^2$

sujeto a

x (k + 1)= x (k) + (k + 1) u (k),
x (0)= 0,
u (k) ∈ ℝ*, para k= 0, 1, 2.*

N= 3.

F (x, u, k)= $u^2 + (x - 2k)^2$.

Final:

$J_3^*\ (x_3)$= S (x (3))
$J_3^*\ (x_3)$= $[x\ (3)]^2$
$J_3^*\ (x_3)$= $[x\ (2) + 3\ u\ (2)]^2$.

Período k= 2:

$J_2^*\ (x_2)$= $\max_{u(2)\in\mathbb{R}} [u\ (2)]^2 + [x\ (2) - 2 * 2]^2 + J_3^*\ (x_3)$
$J_2^*\ (x_2)$= $\max_{u(2)\in\mathbb{R}} [u\ (2)]^2 + [x\ (2) - 4]^2 + [x\ (2) + 3\ u\ (2)]^2$.

$\frac{\partial J_2^*}{\partial u(2)}$= 0
2 u (2) + 2 [x (2) + 3 u (2)] * 3= 0
2 u (2) + 6 [x (2) + 3 u (2)]= 0
2 u (2) + 6 x (2) + 18 u (2)= 0
20 u (2)= -6 x (2)
u (2)= $\frac{-6x(2)}{20}$
u^* (2)= $\frac{-3}{10}$ x (2).

$\frac{\partial^2 J_2^*}{\partial [u(2)]^2}$= 20 > 0.

x (2)= x (1) + 2 u (1).

$J_2^*\ (x_2)$= $[\frac{-3}{10} x\ (2)]^2 + [x\ (2) - 4]^2 + \{x\ (2) + 3\ [\frac{-3}{10} x\ (2)]\}^2$
$J_2^*\ (x_2)$= $\frac{9}{100}\ [x\ (2)]^2 + [x\ (2)]^2$ - 8 x (2) + 16 + $[x\ (2) - \frac{9}{10} x\ (2)]^2$
$J_2^*\ (x_2)$= $\frac{9}{100}\ [x\ (2)]^2 + [x\ (2)]^2$ - 8 x (2) + 16 + $[\frac{1}{10} x\ (2)]^2$
$J_2^*\ (x_2)$= $\frac{9}{100}\ [x\ (2)]^2 + [x\ (2)]^2$ - 8 x (2) + 16 + $\frac{1}{100}\ [x\ (2)]^2$

$J_2^* (x_2)= \frac{11}{10} [x (2)]^2$ - 8 x (2) + 16
$J_2^* (x_2)= \frac{11}{10} [x (1) + 2 u (1)]^2$ - 8 [x (1) + 2 u (1)] + 16
$J_2^* (x_2)= \frac{11}{10} [x (1) + 2 u (1)]^2$ - 8 x (1) - 16 u (1) + 16.

Período k= 1:

$J_1^* (x_1)= \max_{u(1)\in\mathbb{R}} [u (1)]^2 + [x (1) - 2 * 1]^2 + J_2^* (x_2)$
$J_1^* (x_1)= \max_{u(1)\in\mathbb{R}} [u (1)]^2 + [x (1) - 2]^2 + \frac{11}{10} [x (1) + 2 u (1)]^2 - 8 x (1) - 16 u (1) +$ 16.

$\frac{\partial J_1^*}{\partial u(1)}= 0$
2 u (1) + 2 $\frac{11}{10}$ [x (1) + 2 u (1)] * 2 - 16= 0
2 u (1) + $\frac{22}{5}$ [x (1) + 2 u (1)] - 16= 0
2 u (1) + $\frac{22}{5}$ x (1) + $\frac{44}{5}$ u (1) - 16= 0
$\frac{54}{5}$ u (1)= $\frac{-22}{5}$ x (1) + 16
u (1)= $\frac{\frac{-22}{5}x(1)+16}{\frac{54}{5}}$
$u^* (1)= \frac{-11}{27}$ x (1) + $\frac{40}{27}$.

$\frac{\partial^2 J_1^*}{\partial [u(1)]^2}= \frac{54}{5} > 0$.

x (1)= x (0) + u (0)
x (1)= 0 + u (0)
x (1)= u (0).

$J_1^* (x_1)= [u (1)]^2 + [x (1) - 2]^2 + \frac{11}{10} [x (1) + 2 u (1)]^2$ - 8 x (1) - 16 u (1) + 16
$J_1^* (x_1)= [\frac{-11}{27} x (1) + \frac{40}{27}]^2 + [x (1)]^2$ - 2 x (1) + 4 + $\frac{11}{10} \{x (1) + 2 [\frac{-11}{27} x (1) + \frac{40}{27}]\}^2$ - 8 x (1) - 16 $[\frac{-11}{27}$ x (1) + $\frac{40}{27}]$ + 16
$J_1^* (x_1)= \frac{121}{729} [x (1)]^2 - \frac{880}{729}$ x (1) + $\frac{1600}{729} + [x (1)]^2$ - 2 x (1) + 4 + $\frac{11}{10} [x (1) - \frac{22}{27} x (1) + \frac{80}{27}]^2$ - 8 x (1) + $\frac{176}{27}$ x (1) - $\frac{640}{27}$ + 16
$J_1^* (x_1)= \frac{850}{729} [x (1)]^2 - \frac{3428}{729}$ x (1) - $\frac{1100}{729} + \frac{11}{10} [\frac{5}{27} x (1) + \frac{80}{27}]^2$
$J_1^* (x_1)= \frac{850}{729} [x (1)]^2 - \frac{3428}{729}$ x (1) - $\frac{1100}{729} + \frac{11}{10} \{\frac{25}{729} [x (1)]^2 + \frac{800}{729}$ x (1) + $\frac{6400}{729}\}$
$J_1^* (x_1)= \frac{850}{729} [x (1)]^2 - \frac{3428}{729}$ x (1) - $\frac{1100}{729} + \frac{55}{1458} [x (1)]^2 + \frac{880}{729}$ x (1) + $\frac{704}{729}$
$J_1^* (x_1)= \frac{1755}{1458} [x (1)]^2 - \frac{2548}{729}$ x (1) - $\frac{396}{729}$
$J_1^* (x_1)= \frac{1755}{1458} [u (0)]^2 - \frac{2548}{729}$ u (0) + $\frac{396}{729}$.

Período k= 0:

$J_0^*\ (x_0)= \max_{u(0) \in \mathbb{R}} [u\ (0)]^2 + [x\ (0) - 2 * 0]^2 + J_1^*\ (x_1)$
$J_0^*\ (x_0)= \max_{u(0) \in \mathbb{R}} [u\ (0)]^2 - (0 - 0)^2 + \frac{1755}{1458} [u\ (0)]^2 - \frac{2548}{729} u\ (0) + \frac{396}{729}$
$J_0^*\ (x_0)= \max_{u(0) \in \mathbb{R}} \frac{3213}{1458} [u\ (0)]^2 - 0^2 - \frac{2548}{729} u\ (0) + \frac{396}{729}$
$J_0^*\ (x_0)= \max_{u(0) \in \mathbb{R}} \frac{3213}{1458} [u\ (0)]^2 - 0 - \frac{2548}{729} u\ (0) + \frac{396}{729}$
$J_0^*\ (x_0)= \max_{u(0) \in \mathbb{R}} \frac{3213}{1458} [u\ (0)]^2 - \frac{2548}{729} u\ (0) + \frac{396}{729}$.

$\frac{\partial J_0^*}{\partial u(0)}= 0$
$\frac{3213}{729}$ u (0) - $\frac{2548}{729}= 0$
$\frac{3213}{729}$ u (0)= $\frac{2548}{729}$
u (0)= $\frac{\frac{2548}{729}}{\frac{3213}{729}}$
u^* (0)= 0,79.

$\frac{\partial^2 J_0^*}{\partial [u(0)]^2}= \frac{3213}{729} > 0.$

$J_0^*\ (x_0)= \frac{3213}{1458} * 0{,}79^2 + \frac{2548}{729} * 0{,}79 + \frac{396}{729}$
$J_0^*\ (x_0)= \frac{3213}{1458}$ * 0,63 + 2,77 + 0,54
$J_0^*\ (x_0)$= 1,75 + 2,77 + 0,54
$J_0^*\ (x_0)$= 5,06. Valor óptimo del funcional objetivo.

Por lo tanto, se obtienen los siguientes valores óptimos:

$J_0^*\ (x_0)$= 5,06;
u^* (0)= 0,79; u^* (1)= 1,16; u^* (2)= -0,93;
x^* (0)= 0; x^* (1)= 0,79; x^* (2)= 3,11.

Ejercicio 3.

Una mina contiene 80 unidades de mineral. La autoridad responsable de la gestión de la mina debe decidir la cantidad de mineral a extraer en cada uno de los cuatro períodos de que se dispone para su extracción, transcurridos los cuales el valor del mineral que queda en la mina es cero. La cantidad de mineral a extraer en cada uno de los períodos puede ser 0, 10 o 20 unidades, debiendo ser obviamente menor o igual que la cantidad que quede en la mina en el momento de la extracción. El beneficio que se obtiene en un período depende del stock de mineral x que hay en la mina y de la cantidad que se extraiga en dicho período u, según se muestra en la siguiente tabla:

Tabla 1. Beneficio en función del stock de mineral y de la cantidad que se extrae.

x	**0**	**10**	**20**	**30**	**40**	**50**	**60**	**70**	**80**
0	-5	-5	-5	-5	-10	-10	-10	-15	-15
10	-	-35	-25	-20	0	5	15	20	25
20	-	-	-35	-25	-15	-10	10	30	40

Calcular la cantidad de mineral a extraer en cada uno de los cuatro períodos y el correspondiente stock de mineral que va quedando en la mina en cada período.

x (k): stock de mineral que hay en la mina al comienzo del período k + 1.
u (k): cantidad de mineral a extraer de la mina durante el período k + 1.
π (x (k), u (k)): beneficio que se obtiene en un período con stock inicial x (k) y donde la cantidad extraída es u (k).

Problema a resolver:

$$\max_{\{u(k)\}_{k=0}^{3}} J = \sum_{k=0}^{3} \pi \left(x (k), u (k)\right)$$

sujeto a

x (k + 1)= x (k) - u (k),
x (0)= 80,
u (k) $\in$ {0, 10, 20},
u (x) $\leq$ x (k), para k= 0, 1, 2, 3.

N= 4.

F (x, u, k)= π (x (k), u (k)).

<u>Final:</u>

J_4^* (x_4)= S (x (4))
J_4^* (x_4)= 0.

x^* (4)= 40.

<u>Período k= 3:</u>

$J_3^* \ (x_3)= \max\limits_{u(3)\in\{0,10,20\}} \pi \ (x \ (3), u \ (3)) + J_4^* \ (x_4)$
$J_3^* \ (x_3)= \max\limits_{u(3)\in\{0,10,20\}} \pi \ (x \ (3), u \ (3)) + 0$
$J_3^* \ (x_3)= \max\limits_{u(3)\in\{0,10,20\}} \pi \ (x \ (3), u \ (3)).$

x(3)	u(3)	π(x(3),u(3))	x(3)	u(3)	π(x(3),u(3))
80	0	-15	40	0	-10
	10	25		10	0 máximo
	20	40 máximo		20	-15
70	0	-15	30	0	-5 máximo
	10	20		10	-20
	20	30 máximo		20	-25
60	0	-10	20	0	-5 máximo
	10	15 máximo		10	-25
	20	10		20	-35
50	0	-10			
	10	5 máximo			
	20	-10			

x (3)	u^* (3)	J_3^*
80	20	40
70	20	30
60	10	15
50	10	5
40	10	0
30	0	-5
20	0	-5

x^* (3)= 50; u^* (3)= 10.

<u>Período k= 2:</u>

$J_2^* \ (x_2)= \max\limits_{u(2)\in\{0,10,20\}} \pi \ (x \ (2), u \ (2)) + J_3^* \ (x_3).$

x(2)	u(2)	$\pi(x(2),u(2))+J_3^*$	x(2)	u(2)	$\pi(x(2),u(2))+J_3^*$
80	0	$-15+40=25$	50	0	$-10+5=-5$
	10	$25+30=55$ máximo		10	$5+0=5$ máximo
	20	$40+15=55$ máximo		20	$-10+(-5)=-15$
70	0	$-15+30=45$	40	0	$-10+0=-10$
	10	$20+15=35$ máximo		10	$0+(-5)=-5$ máximo
	20	$30+5=35$ máximo		20	$-15+(-5)=-20$
60	0	$-10+15=5$			
	10	$15+5=20$ máximo			
	20	$10+0=10$			

x (2)	u^* **(2)**	J_2^*
80	10 ó 20	55
70	10 ó 20	35
60	10	20
50	10	5
40	10	-5

x^* (2)= 60; u^* (2)= 10.

<u>Período k= 1:</u>

$J_1^*(x_1)=\max_{u(1)\in\{0,10,20\}} \pi\,(x\,(1),u\,(1))+J_2^*\,(x_2).$

x(1)	u(1)	$\pi(x(1),u(1))+J_2^*$
80	0	$-15+55=40$
	10	$25+35=60$ máximo
	20	$40+20=60$ máximo
70	0	$-15+35=20$
	10	$20+20=40$ máximo
	20	$30+5=35$
60	0	$-10+20=10$
	10	$15+5=20$ máximo
	20	$10+(-5)=5$

x (1)	$\boldsymbol{u^*}$ **(1)**	$\boldsymbol{J_1^*}$
80	10 ó 20	60
70	10	40
60	10	20

x^* (1)= 70; u^* (1)= 10.

<u>Período k= 0:</u>

J_0^* (x_0)= $\max\limits_{u(0)\in\{0,10,20\}} \pi\ (x\ (0), u\ (0)) + J_1^*\ (x_1)$
J_0^* (x_0)= $\max\limits_{u(0)\in\{0,10,20\}} \pi\ (x\ (0), u\ (0)) + J_1^*\ (x_0 - u_0)$
J_0^* (x_0)= $\max\limits_{u(0)\in\{0,10,20\}} \pi\ (x\ (0), u\ (0)) + J_1^*\ (80 - u_0)$.

x(0)	u(0)	$\pi(x(0), u(0)) + J_1^*$
80	0	$-15 + 60 = 45$
	10	$25 + 40 = 65$ máximo
	20	$40 + 20 = 60$

x (0)	$\boldsymbol{u^*}$ **(0)**	$\boldsymbol{J_0^*}$
80	10	65

x^* (0)= 80; u^* (0)= 10; J_0^* (x_0)= 65.

Haciendo un recorrido de principio a final, se tiene que x^* (0)= 80 y:

u^* (0)= 10; x^* (1)= x^* (0) - u^* (0)= 70.
u^* (1)= 10; x^* (2)= x^* (1) - u^* (1)= 60.
u^* (2)= 10; x^* (3)= x^* (2) - u^* (2)= 50.
u^* (3)= 10; x^* (4)= x^* (3) - u^* (3)= 40.

Por lo tanto, la cantidad de mineral a extraer en cada uno de los cuatro períodos es 10 y el correspondiente stock de mineral que va quedando en la mina en cada período es 70, 60, 50 y 40, respectivamente.